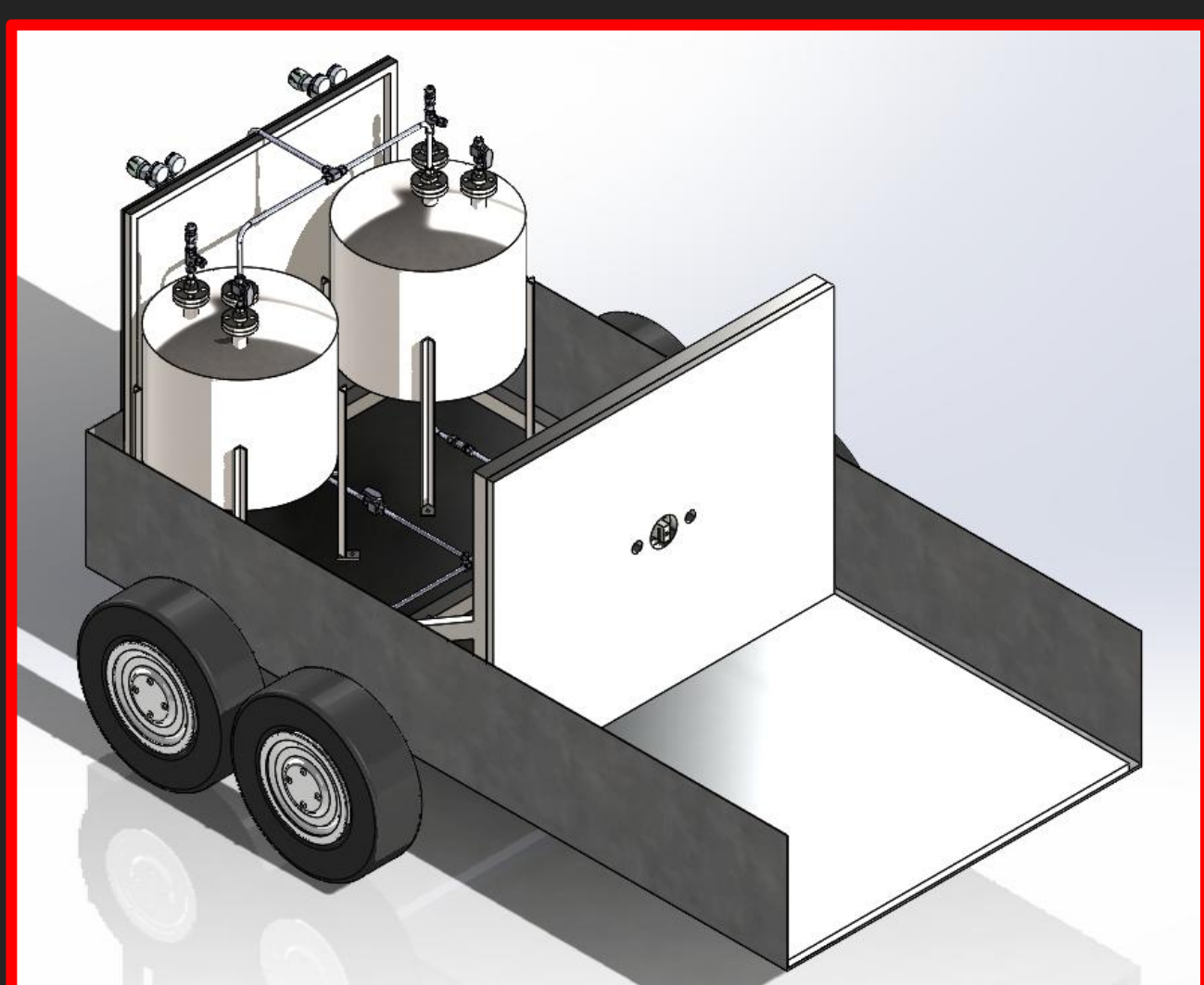




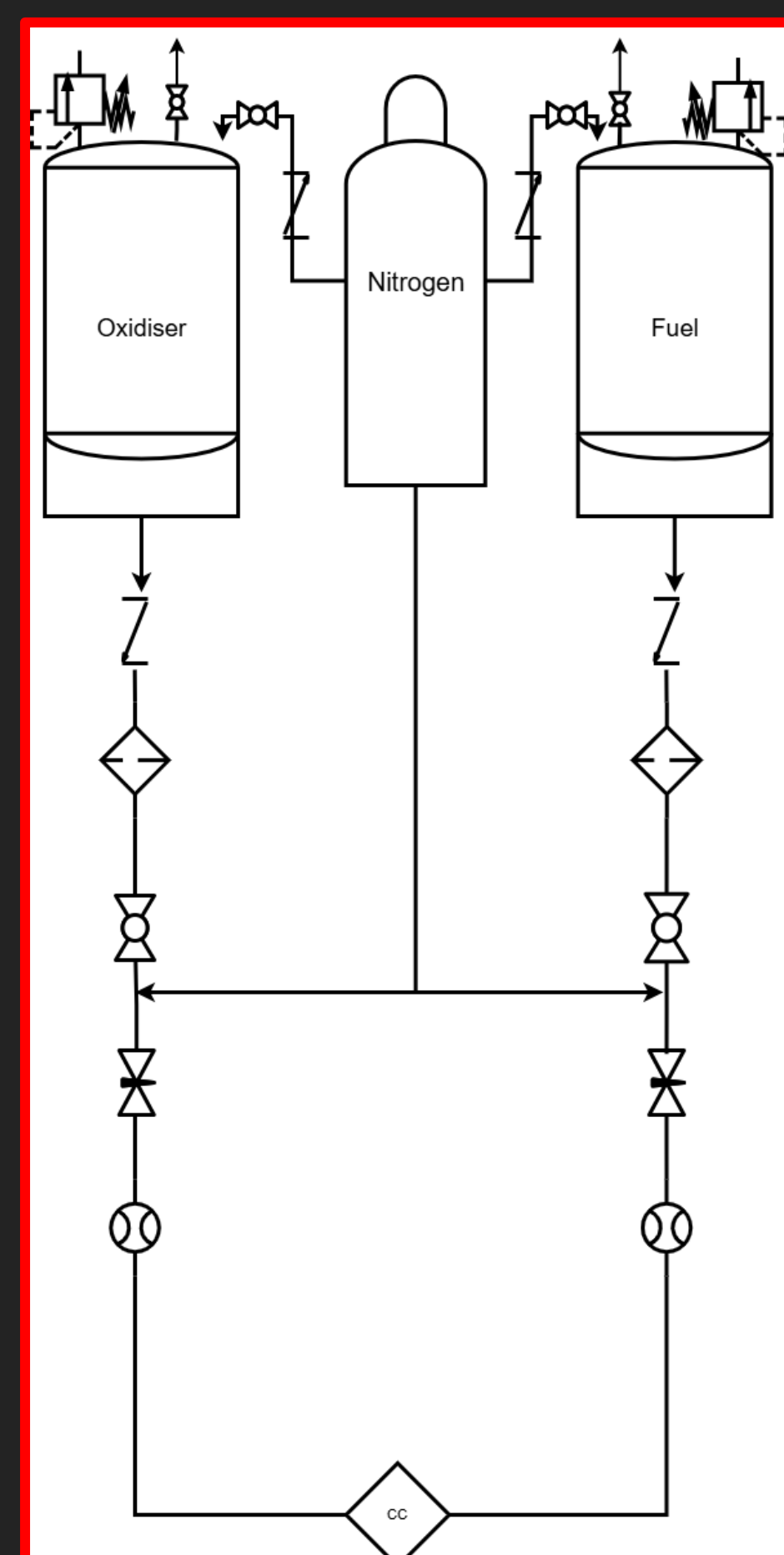
Development of a Mobile Rocket Test Stand on a Trailer

DAIRE JOSEPH O'SULLIVAN

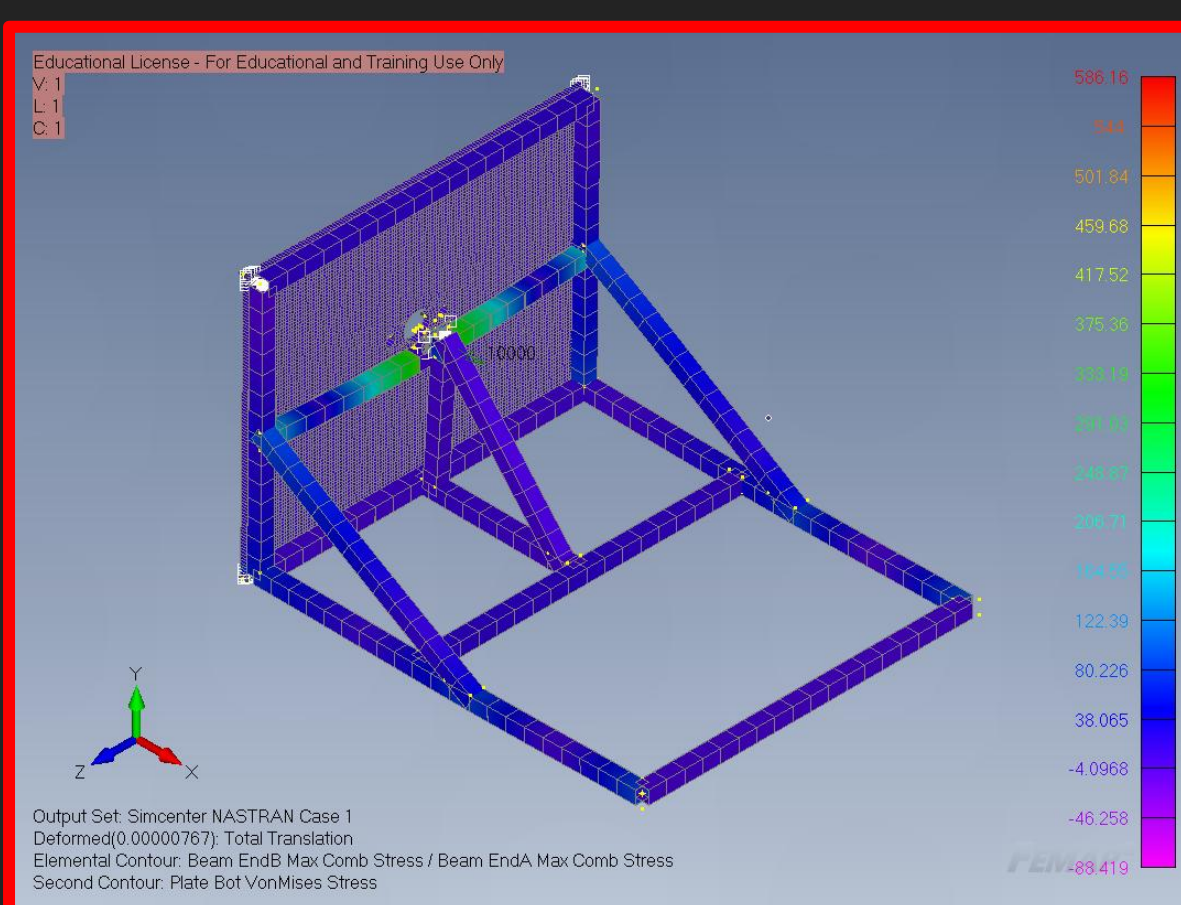
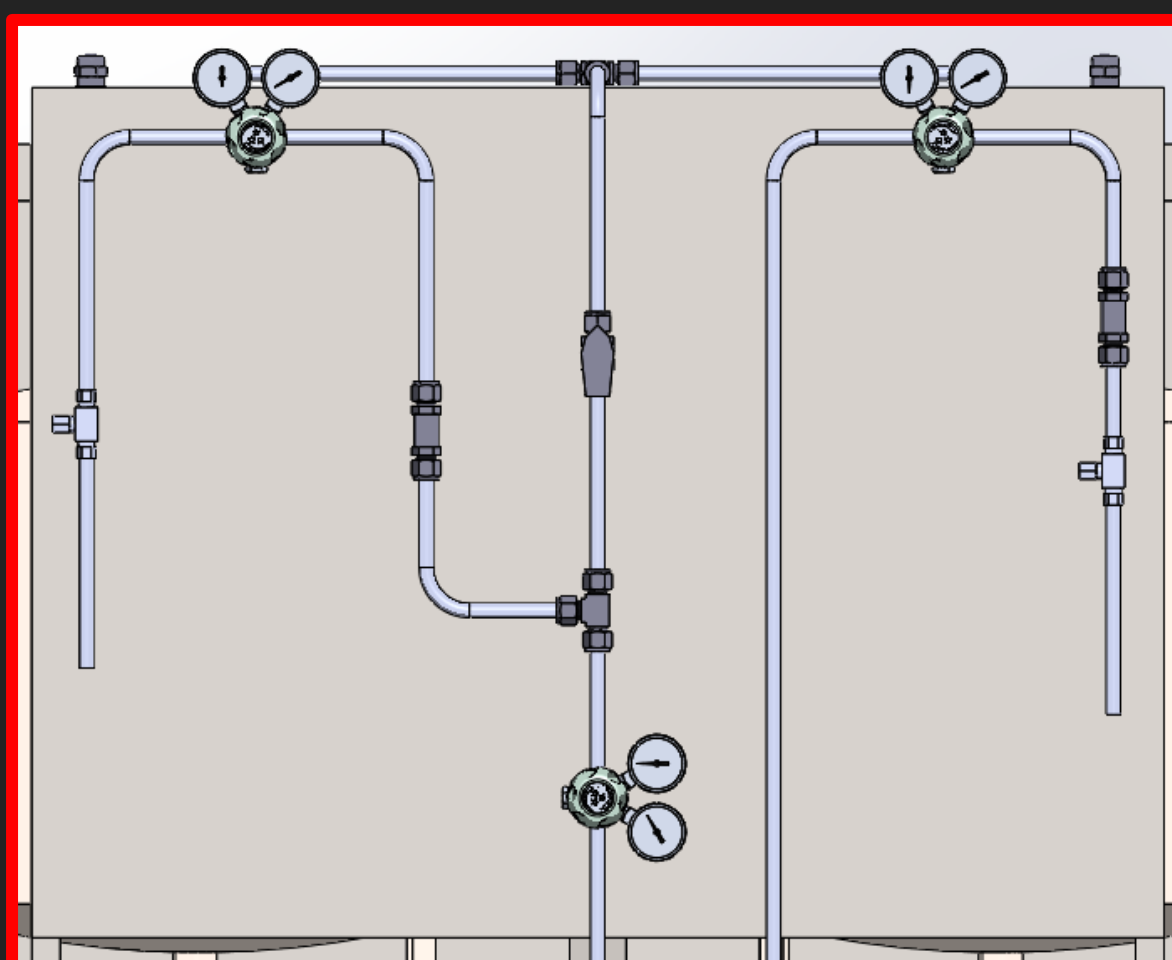
There is an increase in rocket propulsion development on university campuses driven both by academic research and student-led extracurricular groups. Independent and easy access to test facilities allows for quicker prototyping and iteration and a reduction in costs associated with using external test stands. Specifically, a mobile platform enables testing to take place in remote areas thereby reducing safety risks and noise pollution in populated areas. This research is focused on the design of a modular, mobile test stand while also outlining the considerations and methods needed to develop testing facilities.



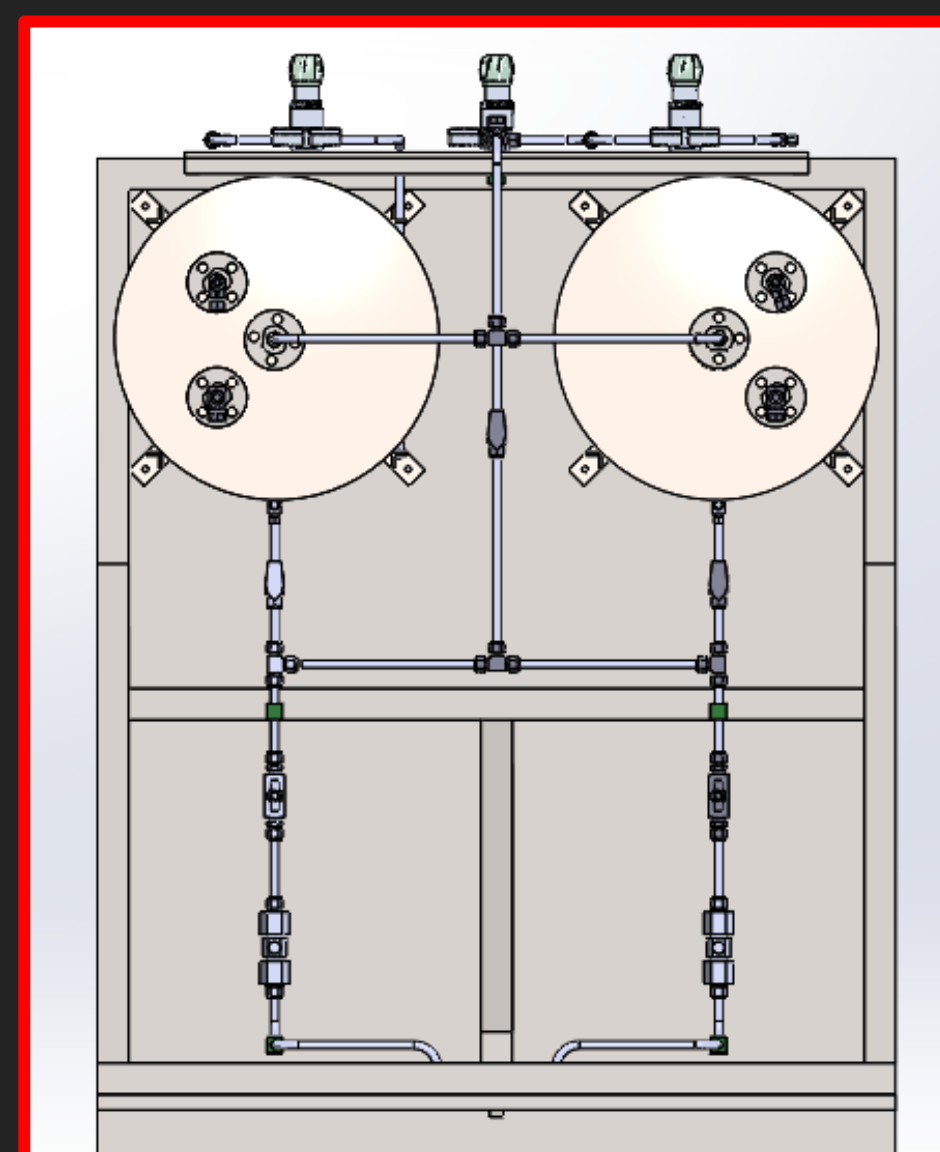
- There are three major elements:
 - Fluid delivery
 - Thrust structure
 - Data Acquisition and Control
- These will vary depending on engine type and project specific requirements.



- The size of tanks and valves increase with thrust requirements which means the fluid system can be the limiting factor on small trailers.
- Several methods can be used to restrain the trailer during tests including sandbags, wheel chocks and ground anchors.



- For student groups, many options exist to reduce cost and complexity such as:
 - Lower fidelity sensors.
 - Repurposing consumer electronics .
 - Limiting test flexibility.



This design will be built on a 1.3m x 2.6m twin-axle trailer and is designed to enable the testing of engines up to 10kN of thrust with nitrous oxide and hydrocarbon propellants pressurised up to 10MPa with nitrogen gas. A metering valve allows for the variation of flow rates. The system is designed to be operated completely remotely using electrically actuated valves. Data on the thrust, pressure and temperature of the engine will be gathered.

